

ABDUL MUQTADIR

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EDUCATION

National University of Sciences and Technology (NUST)

Bachelor of Engineering in Avionics Engineering; CGPA: 3.73 / 4.00 (Semester 7)

Sep 2023 – Expected Jun 2027

Risalpur, Pakistan

- Honors & Coursework:** Dean's Honor List across 7 consecutive semesters. Coursework in RF & Microwave Design, Antenna & Wave Propagation, Digital Signal Processing (DSP), Control Systems Engineering, and Microcontrollers.

The Universal Millennium College

Cambridge International A-Levels — Mathematics (A), Physics (A), Chemistry (A)

Aug 2021 – May 2023

Pakistan

TECHNICAL SKILLS

RF & Antennas: CST Studio Suite 2025 (3D Full-Wave EM), Keysight ADS, LineCalc, S-Parameters (S11, S21 return/insertion loss), CPW & Microstrip matching, Super Wideband Vivaldi Antennas, Conformal Arrays.

Communications & SDR: ADALM-Pluto Software-Defined Radio, 915 MHz FM Telemetry, Quadrature Demodulation, Digital IIR/FIR Filter Design, Noise Suppression.

Controls & Modeling: Closed-Loop Digital PID Control, Linear Quadratic Regulator (LQR), State-Space Modeling, IMU 6-DOF Sensor Fusion (Kalman & Complementary Filters), MATLAB / Simulink.

Programming & Firmware: C, C++, Embedded C, Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, G-Code, Linux/Bash, Git.

Hardware & Prototyping: ESP32 Dual-Core, STM32, Arduino Uno, MPU6050 IMU, BTS7960 43A Motor Drivers, Stepper Motors, CATIA V5, AutoCAD, KiCAD, Oscilloscope Diagnostics.

SELECTED ENGINEERING PROJECTS

Super Wideband (3–18 GHz) Conformal Vivaldi Antenna

Aug 2026

CST Studio Suite 2025, Rogers RT6035HTC, CPW Transition

NUST CAE

- Designed and simulated an ultra-compact planar and conformal Vivaldi antenna achieving a 6:1 impedance bandwidth ratio with return loss $S_{11} < -10$ dB across the entire 3.0 to 18.0 GHz band for airborne Electronic Warfare (EW).
- Synthesized a single-layer Co-Planar Waveguide (CPW) transition and exponential flare (opening factor 0.16 mm^{-1}) with 4 curved ground slits to eliminate destructive higher-order cross-polarization modes.
- Modeled aerodynamic conformal bending to match UAV fuselage curvature and validated inter-element mutual coupling isolation ($S_{21} < -25$ dB).

Hardware-in-the-Loop (HIL) UAV Altitude Control & SDR Telemetry

Jul 2026

ADALM-Pluto SDR, Python Dynamics Plant, Closed-Loop PID Autopilot

NUST CAE

- Constructed a real-time HIL simulation coupling a simulated second-order UAV climb dynamics plant with a closed-loop digital PID autopilot.
- Modulated continuous altitude telemetry over a physical 915 MHz RF link using an ADALM-Pluto SDR; built receiver DSP pipeline with quadrature demodulation and single-pole IIR noise filtering to reconstruct the control feedback loop under noise.

2.4 GHz 5th-Order Chebyshev Stepped-Impedance Low-Pass Filter

Jun 2026

Keysight ADS, Keysight LineCalc, CST Microwave Studio, FR4 Microstrip

NUST CAE

- Synthesized a distributed stepped-impedance microstrip low-pass filter at a 2.4 GHz cutoff frequency for avionics ISM-band harmonic suppression.
- Conducted Monte Carlo statistical yield analysis to model PCB substrate tolerance variations (dielectric constant $\epsilon_r = 4.4 \pm 0.1$) and verified 3D electromagnetic S-parameters in CST.

Two-Wheeled Inverted Pendulum Robot: LQR vs. Cascaded PID

May 2026

MATLAB/Simulink, Embedded C++, MPU6050 6-DOF IMU

NUST CAE

- Derived non-linear Lagrangian equations of motion and linearized state-space matrices around the unstable vertical equilibrium.
- Demonstrated physical degradation of theoretical LQR under unobserved linear cart states; engineered and stabilized a physical prototype via cascaded PID firmware with anti-windup clamping and complementary sensor filtering.

EXPERIENCE & LEADERSHIP

Freelance Engineering Consultant

Jul 2025 – Present

Independent Contractor — Embedded Systems, Firmware & RF

Remote

- Consulted on microcontroller firmware architecture, I2C/SPI sensor interfaces, and closed-loop control algorithms for international engineering clients.
- Conducted schematic reviews, circuit debugging, and component selection for high-reliability embedded hardware builds.

Head of Avionics Club

Sep 2024 – Present

College of Aeronautical Engineering, NUST

Risalpur, Pakistan

- Lead technical student projects, SDR telemetry experiments, and autonomous robotics workshops for 40+ engineering students.

A-Level STEM Tutor (Mathematics & Physics)

Nov 2022 – Present

Private Mentorship

Remote & In-Person

- Mentored 20+ students in differential calculus, kinematics, wave mechanics, and electromagnetism, with students achieving top A/A* grades.